Aleksandar Armacki

Contact

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Research Interests

I study what performance guarantees are achievable in large-scale single- and multi-agent learning systems under challenges such as heavy-tailed noise, data heterogeneity and limited communication. To that end, I focus on exploiting problem-specific structure to obtain sharp guarantees and design algorithms that attain them. My work spans the areas of stochastic optimization and machine learning, utilizing tools such as concentration inequalities, large deviation techniques and consensus theory.

Experience

Sep 2025 - Present	Post-doctoral researcher at EPFL, Lausanne, Switzerland *Adaptive Systems Laboratory* | Advisor: Prof. **Ali H. Sayed**
Aug 2020 - Aug 2025	Research assistant at Carnegie Mellon University, Pittsburgh, PA *Electrical and Computer Engineering* | Advisor: Prof. **Soummya Kar**
Sep 2019 - Dec 2019	Visiting researcher at University of Pennsylvania, Philadelphia, PA *Electrical and Systems Engineering* | Advisor: Prof. **Alejandro Ribeiro**
Nov 2018 - Jun 2020	Junior researcher at Faculty of Sciences, Novi Sad, Serbia *Numerical Analysis Group* | Advisor: Prof. **Dušan Jakovetić**
Jul 2018 - Oct 2018	Junior researcher at BioSense Institute, Novi Sad, Serbia
Feb 2018 - Jul 2018	Visiting researcher at FRI, Ljubljana, Slovenia *Bioinformatics Lab* | Advisor: Prof. **Blaž Zupan**

Education

Aug 2020 - Aug 2025	Doctor of Philosophy in Electrical and Computer Engineering College of Engineering, Carnegie Mellon University Thesis: "High-probability and Large Deviations Techniques for Design and Analysis of Large-scale and Distributed Learning Systems". [Link to thesis]
Oct 2016 - Oct 2018	Master of Science in Applied Mathematics Faculty of Sciences, University of Novi Sad Thesis: "Application of Autoencoders on Single-cell Data". [Link to thesis]
Sep 2012 - July 2016	Bachelor of Science in Applied Mathematics Faculty of Sciences, University of Novi Sad

# PUBLICATIONS (IN REVERSE CHRONOLOGICAL ORDER, * = EQUAL CONTRIBUTION)

Journal Articles

[J1] **Armacki, A.**, Bajović, D., Jakovetić, D., Kar, S. (2026). *Sharp High-Probability Rates for Nonlinear SGD under Heavy-Tailed Noise via Symmetrization*. IEEE Transactions on Information Theory. [Published version] [Preprint]

[J2] **Armacki, A.***, Yu, S.*, Bajović, D., Jakovetić, D., Kar, S. (2026). *Large Deviation Upper Bounds and Improved MSE Rates of Nonlinear SGD: Heavy-Tailed Noise and Power of Symmetry*. SIAM Journal on Optimization. [Published version] [Preprint]

[J3] **Armacki, A.**, Bajović, D., Jakovetić, D., Kar, S. (2025). *Distributed Center-Based Clustering: A Unified Framework*. IEEE Transactions on Signal Processing. [Published version] [Preprint]

[J4] **Armacki, A.**, Bajović, D., Jakovetić, D., Kar, S. (2024). *A One-shot Framework for Distributed Clustered Learning in Heterogeneous Environments*. IEEE Transactions on Signal Processing. [Published version] [Preprint]

Conference Papers

[C1] **Armacki, A.**, Bajović, D., Jakovetić, D., Kar, S., Sayed, A. H. (2026). *Tight Long-Term Tail Decay of (Clipped) SGD in Non-Convex Optimization*. 39th Conference on Learning Theory (COLT). [Published version] [Preprint]

[C2] **Armacki, A.**, Sayed, A. H. (2026). *High-Probability Convergence Guarantees of Decentralized SGD*. 43rd International Conference on Machine Learning (ICML). [Preprint]

[C3] **Armacki, A.**, Kar, S. (2025). *Toward Understanding the Improved Robustness to Initialization in Distributed Clustering*. 33rd European Signal Processing Conference. [Published version] [Preprint]

[C4] **Armacki, A.**, Yu, S., Sharma, P., Joshi, G., Bajović, D., Jakovetić, D., Kar, S. (2025). *High-probability Convergence Bounds for Online Nonlinear Stochastic Gradient Descent under Heavy-tailed Noise*. 28th International Conference on Artificial Intelligence and Statistics (AISTATS). [Published version] [Preprint]

[C5] **Armacki, A.***, Sharma, H.*, Bajović, D., Jakovetić, D., Chakraborty, M., Kar, S. (2024). *Distributed Gradient Clustering: Convergence and the Effect of Initialization*. 58th Asilomar Conference on Signals, Systems, and Computers. [Published version] [Preprint]

[C6] **Armacki, A.**, Milosevic, N., Bajović, D., Kar, S., Jakovetić, D., Bakhtiarnia, A., Esterle, L., Muscat, A., Festi, T. (2023). *Communication Efficient Model-aware Federated Learning for Visual Crowd Counting and Density Estimation in Smart Cities*. 31st European Signal Processing Conference. [Published version] [Preprint]

[C7] **Armacki, A.**, Bajović, D., Jakovetić, D., Kar, S. (2022). *Personalized Federated Learning via Convex Clustering*. IEEE International Smart Cities Conference (ISC2). [Published version] [Preprint]

[C8] **Armacki, A.**, Bajović, D., Jakovetić, D., Kar, S. (2022). *Gradient Based Clustering*. 39th International Conference on Machine Learning (ICML). [Published version] [Preprint]

[C9] **Armacki, A.**, Jakovetić, D., Krejić, N., Krklec Jerinkić, N. (2019). *Distributed Trust-Region Method With First Order Models*. IEEE EUROCON 18th International Conference on Smart Technologies. [Published version] [Preprint]

Books and Textbooks

[B1] Jakovetić, D., **Armacki, A.** (2020). *Distributed Optimization with Applications*. E-textbook. Faculty of Sciences, University of Novi Sad. [Link to textbook]

Preprints

[P1] **Armacki, A.**, Cai, H., Sayed, A. H. (2026). *Decentralized SGD under Heavy-Tailed Noise: Optimal Convergence Rates and the Role of Gradient Clipping*. Under review. [Preprint]

[P2] Cai, H., Zhao, Y., **Armacki, A.**, Chen, J., Sayed, A. H. (2026). *A Unified Zeroth-Order Approach for Decentralized Minimax Optimization*. arXiv preprint. [Preprint]

[P3] **Armacki, A.**, Cai, H., Sayed, A. H. (2026). *High-Probability Convergence in Decentralized Stochastic Optimization with Gradient Tracking*. Under review. [Preprint]

Awards and Honors

2026 ICML Gold Reviewer - Awarded to top 25% of reviewers based on Area Chair ratings.
2023 IBM PhD Fellowship - Finalist (one of three nominees by the ECE Department at CMU).
2020 Dean's Fellowship - ECE Department, Carnegie Mellon University.
2019 Annual Award of the Mathematical Institute of the Serbian Academy of Sciences and Arts in the Field of Computing for M.Sc. Students - Finalist.

Mentoring and Teaching

- *Student mentoring:*
  1. **Haoyuan Cai** - PhD student, EPFL (Fall 2025 - Present). Helping supervise and collaborating with since joining Prof. Sayed's group. The supervision resulted in works [P1,P2,P3].
  2. **Himkant Sharma** - Research intern, CMU (Summer 2024). supervised for the duration of the internship in Prof. Kar's group. The supervision resulted in publication [C5].
- *Teaching:* Served as teaching assistant for the graduate course "Linear Systems" in 2022 and 2023, at the ECE Department, CMU. Responsibilities included conducting recitations, holding office hours, and creating homework assignments, midterm and final exams.

Academic Service

- *Conference reviewer:* IEEE ISIT, AISTATS, ICML, NeurIPS, AAAI.
- *Journal reviewer:* IEEE Transactions on Signal Processing, Journal of Machine Learning Research.
- *Student council for faculty hiring:* Served on the ECE Department's student council for faculty hiring in 2023. Participated in interviewing faculty candidates to assess their potential to effectively mentor graduate students and foster healthy advisor-mentee relationships.

Talks and Presentations

2026 *Invited talk, "Bogoljub Stanković" Seminar*, University of Novi Sad, Serbia
2026 *43rd International Conference on Machine Learning*, Seoul, South Korea
2026 *39th Conference on Learning Theory*, San Diego, USA
2025 *33rd European Signal Processing Conference*, Palermo, Italy
2025 *28th International Conference on Artificial Intelligence and Statistics*, Mai Khao, Thailand
2024 *58th Asilomar Conference on Signals, Systems, and Computers*, Monterey Bay, USA
2023 *31st European Signal Processing Conference*, Helsinki, Finland
2022 *39th International Conference on Machine Learning*, Baltimore, USA
2019 *IEEE EUROCON 18th International Conference on Smart Technologies*, Novi Sad, Serbia
2018 *US-Serbia & West Balkan Data Science Workshop*, NSF Serbia, Belgrade, Serbia